

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Seventh Semester B.E. (ISE) (Credit System) Degree Examinations**

January 2023

19IS702 – INFORMATION AND NETWORK SECURITY

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I

	Marks	BT*	CO*	PO*
1. a) How encryption take place in Play fair cipher? illustrate the same by considering, Plaintext: "communicate" and key "COMPUTER".	6	L*2	1	1
b) With an example, illustrate encryption and decryption in Hill Cipher. Also discuss its security.	8	L2	1	1
c) State the motivation for the feistel structure. Describe in brief Block cipher design principles.	6	L2	1	1
2. a) Using Vigenere' cipher demonstrate the encryption and decryption for the given Plaintext: "javapoint" and keyword: "BEST"	6	L2	1	1
b) What is avalanche effect? With a neat diagram, explain DES encryption and decryption process.	8	L2	1	1
c) Discuss the applications and requirements for Public key crypto systems.	6	L2	2	1
3. a) A RSA cryptosystem uses two prime numbers 3 and 13 to generate the public key= 3 and the private key = 7. What is the value of cipher text for a plain text, 5? Also discuss any four possible approaches to attacking the RSA algorithm.	8	L2	2	1
b) In a Diffie-Hellman Key Exchange, Alice and Bob have chosen prime value $q = 17$ and primitive root = 5. If Alice's secret key is 4 and Bob's secret key is 6, what is the secret key they exchanged?	4	L2	2	1
c) With an example, explain Elliptic Curve Encryption and Decryption.	8	L2	2	1

Unit – II

4. a) Explain working of automatic key distribution for connection oriented protocol with a neat diagram.	10	L2	3	1
b) With a neat diagram, give the format of X-509 certificate.	10	L2	3	1
5. a) What is Kerberos? With a neat diagram, explain the overview of Kerberos.	10	L2	3	1
b) Explain the approach used for one way authentication using asymmetric encryption providing confidentiality as well as authentication.	10	L2	4	1
6. a) Explain the SSL architecture and SSL record protocol.	10	L2	4	1
b) With a diagram, explain SSH transport layer protocol stack exchanges along with the packet formation.	10	L2	4	1

Unit – III

7. a) Explain IEEE 802.11i phases of operation in brief.	10	L2	5	1
b) Briefly explain the MIME and S/MIME functionalities.	10	L2	5	1
8. a) Explain steps involved in transmission and reception of PGP messages along with diagram.	10	L2	5	1
b) Compare transport mode versus tunnel mode encryption. With a diagram, describe the scope of ESP encryption and authentication.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
